

20250520EB_MN (NCT,ATP,TDP43)

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2025年5月20日 星期二

Differentiation: BP w/ 0.5mM pyruvate (no extra pyruvate). Cells on glass didn't look very well.

Experiment:

- 2.5mM and 0.15mM: T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A + glutamax, b-mercaptoethanol, N2, B27)
- 0mM: Mix of the following
 - T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A + glutamax, b-mercaptoethanol, N2, B27)
 - T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A + N2, B27)

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年5月22日 星期四

T2 -> T3

2025年5月24日 星期六

T3 -> T4 (~14 mL for over-weekend)

2025年5月27日 星期二

T4 -> T5

2025年5月29日 星期四

T5 -> T6

2025年5月31日 星期六

T6 -> T6 (~14mL for over-weekend)

2025年6月3日 星期二

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.
6. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube (use 70um strainer this time)
10. Spin the cell down at 300rcf, 7min. Remove the supernatent.
11. Add T7 (~5mL) to resuspend the cells and count it.
12. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info				^
	A	B	C	
1	cell	passage	well#	
2	WT11	16+11		6

Table1			^
	A	B	
1	A. a half of cells	volume (mL)	
2	accutase		2
3	HI-FBS/CTWM		4
4	B. a half of cells		
5	0.25% trypsin		2
6	HI-FBS/CTWM		4

Table3											
	A	B	C	D	E	F	G	H	I	J	K
1		conc. (M/mL)	survival	total volume	tota cell count	cell count in total	plate#	well plate	cell count/well (K)	number of wells	volume
2	A	1.98	94	3	5.94	10.59	A	24	252	24	6
3	B	1.55	99	3	4.65		B	24	252	18	5
4											

PlateA							^
	1	2	3	4	5	6	
A	111		92		no virus		
B							
C	110						
D							

PlateB (for Sydney)							^
	1	2	3	4	5	6	
A							
B							
C							
D							

- 2025年6月4日 星期三
- Change T7 -> T7
- 2025年6月5日 星期四
- 1/2 BP T8 -> 1/2 BP T8
- 2025年6月7日 星期六
- 1/2 BP T8 -> 1/2 BP T8
- 2025年6月9日 星期一
- 1/2 BP T8 -> 1/2 BP T8
- 2025年6月11日 星期三
- ☒ 1/2 T8 -> 1/2 T8 (CD56)

Imaging plate A

***Note:**

I prepared new no glucose basal media. I only added P/S, arginine, and lysine into the basal media. Without Glutamax, B-mercaptoethanol, N2, B27 w/o vitA in stock.

For the T8 no glucose, I added T8 supplements (AA, RA, DAPT, BDNF, GDNF), N2, B27. However, I didn't add Glutamax and B-mercaptoethanol

For all the 2.5mM glucose and 0.15 mM glucose below, they are prepared previously and with Glutamax and B-mercaptoethanol. Since no glucose media is not enough, I mixed new and old media together (I forgot they are different), so the 0mM glucose group has fewer Glutamax and B-mercaptoethanol

vLH92 40x

vLH92 60x

TDP43 1h

TDP43 24h

PlateA_conditions							^
	1	2	3	4	5	6	
A	111 (cells looked bad)				1	2	
B			1	2	4	3	
C	110 (cell looked bad)		4	3	1	2	
D			1	3	4	3	

Table5			^
	A	B	
1	number	condition	
2		1 2.5mM	
3		2 0.15mM	
4		3 0mM	
5		4 0mM+FCCP+2DG	

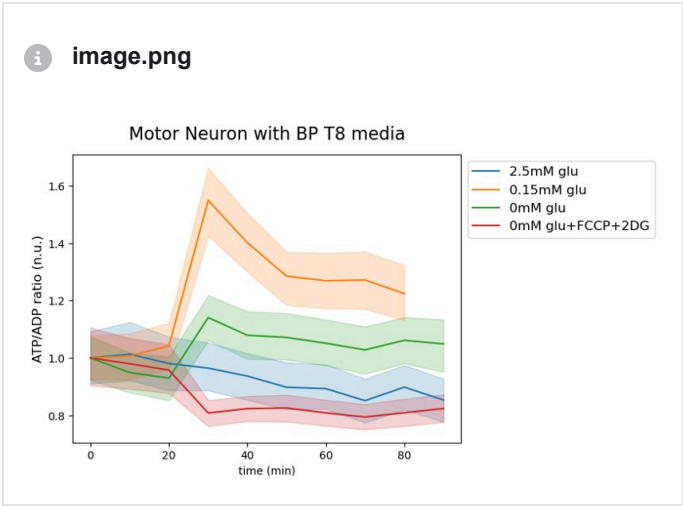
vLH92 (compare 40x and 60x)

(Stephanie picked the point for the 2-4 conditions of 40x and all points of 60x)

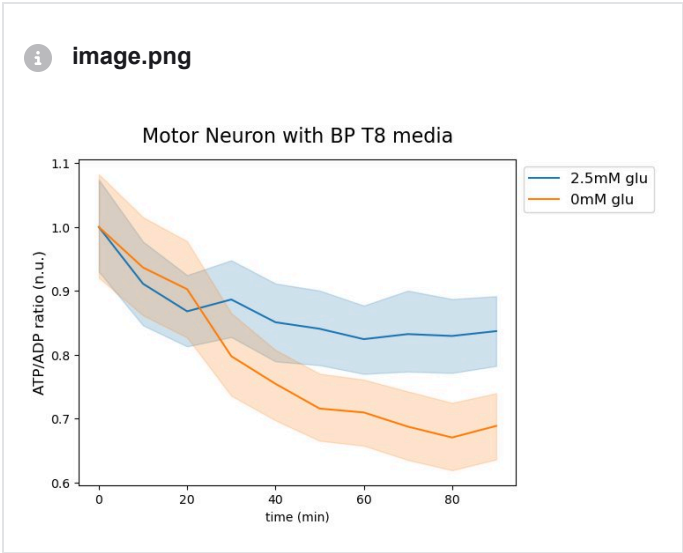
O.C. , 40X, zoom1.5, 600x900, 16-bit

	A	B	C	D
1	channel	bit	power (%)	exposure time
2	488	16	25	300
3	Dapi-ex-Green-em	16	20	500

analysis: the result for 40x is weird... I have never seen this before. I didn't see any analysitcal problem so far
potential reasons: Stephanie picked the cells in the middle of the condition 2 to a part of condition 4. She said she shined the cells for longer time, so they might have photo-toxicity already and some of them died at the end.



60x seems normal (But laser power might be too strong)



TDP43 (AB56)

Treat all cells around 7pm and find positions all together.

Take images around 8:10-8:30

Table4							
	A	B	C	D	E	F	G
1	condition #	glucose conc. (mM)	volume	Halo ligand (2.5uM -> 1nM)	SPY595 (2500x)	FCCP	2DG
2	1	2.5	1000	0.4	0.4		
3	2	0.15	1000	0.4	0.4		
4	3	0	1000	0.4	0.4		
5	4	0mM+FCCP+2 DG	1000	0.4	0.4	1	3.0

O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	500ms	
3	640	16	20	500ms	
4	640 movie	16	40	500ms	40 loops

2025年6月12日 星期四

☒ Treat cells at 11:05 am (CD56)

2025年6月13日 星期五

Image Plate A (C5, D56 all died, AB56 look sick, so I didn't do the experiment)

Hypothesis: it might because I add more Na Pyruvate to them on Wednesday and change back to the one with lower Na pyruvate?